Computer Vision Engineer Task: Egg Segmentation and Video Tracking

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Project Repository: https://github.com/kushagra9453/egg-segmentation

Tracked Video Deliverable (Google Drive): https://drive.google.com/file/d/1bmdephNfQXLSc2-urg9mRzZL_sNBW8Ap/view?usp=drive_link

Executive Summary

This report presents an end-to-end classical computer vision and tracking solution for segmenting and tracking eggs across diverse, visually complex real-world scenes. The project comprises four core tasks:

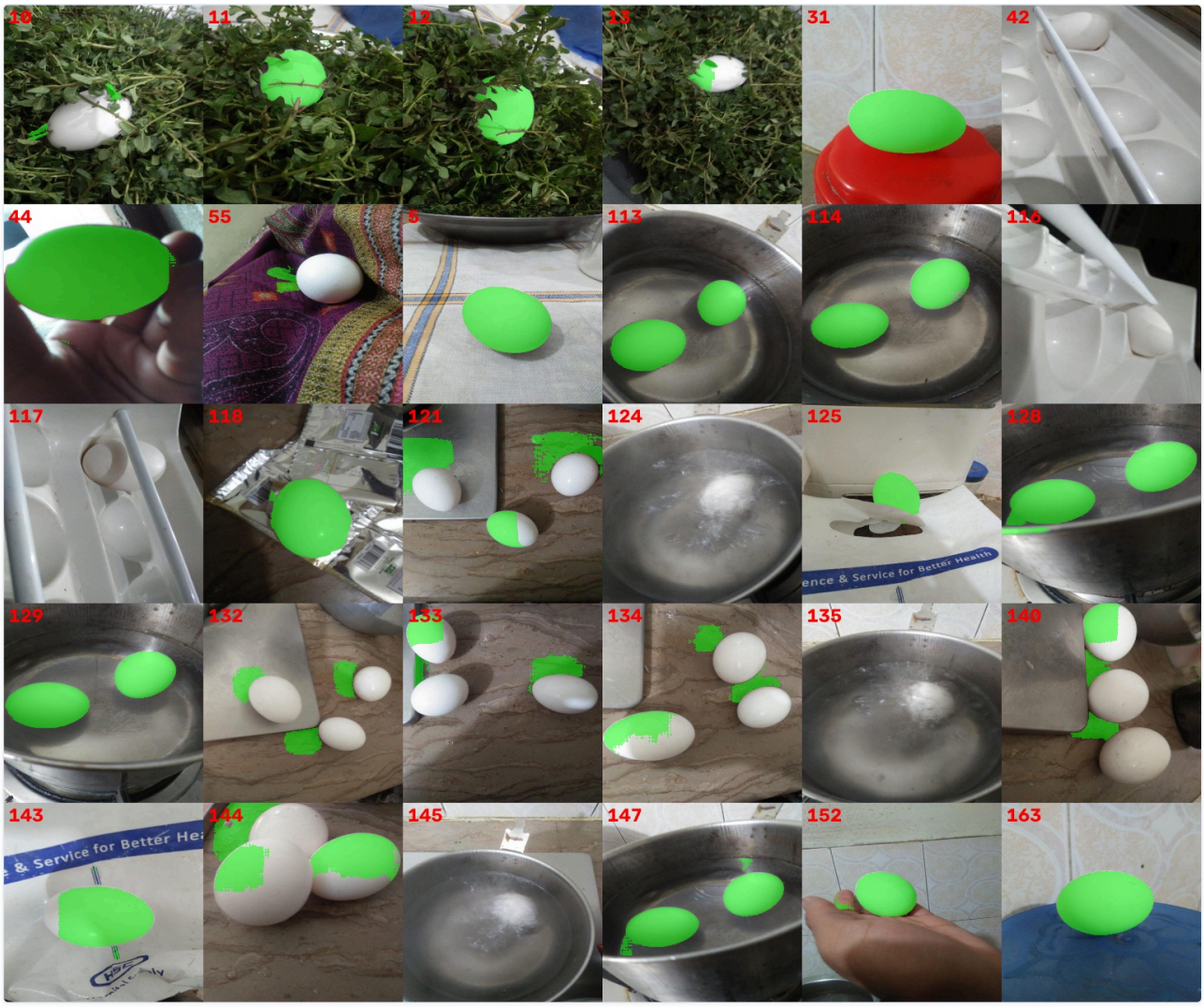
1. **Task 1: Ground-Truth Generation**: Constructed pseudo ground-truth masks for 110 images using Meta's Segment Anything Model (SAM), enhanced with manual bounding boxes, CVAT polygon corrections, and parametric ellipse fitting.
2. **Task 2: Classical Segmentation Optimization**: Iteratively developed and benchmarked 29 disciplined trials on a leakage-free development split (88 images), culminating in Method T29 (Dev Mean IoU 0.773, Boundary F1 0.670).
3. **Task 3: Evaluation & Error Analysis**: Evaluated generalizability on an unseen evaluation split (22 images) using IoU, Dice, Precision, Recall, and Boundary F-score metrics, accompanied by scene-stratified analysis and visual error diagnosis for the worst failures (including the two 0.000 eval images).
4. **Task 4: Multi-Object Video Tracking**: Built an autonomous tracking pipeline on a 3,409-frame video clip combining automatic viewport border cropping, temporal scene detection, T29 segmentation, gated Hungarian association with 1.5 s occlusion memory, and ghost-box elimination.

1. Task 1: Ground-Truth Generation

1.1 Methodology

High-quality pixel-level annotations are essential for benchmark validity. Pseudo ground-truth masks were generated using **Meta's Segment Anything Model (SAM)** (`vit_b` and `vit_h` checkpoints) followed by multi-stage refinement:

1. **Automated Point Grid Prompting (`02_sam_segment.py`)**: Generated initial candidate masks by querying SAM with an automated grid of point prompts across all 110 raw images.
2. **Heuristic Pruning & Filtering (`03_fix_and_refine.py`)**: Filtered raw SAM masks by retaining components with area $\ge 35\%$ of the largest candidate, color consistency in CIELAB space ($\Delta E \le 25$), and non-overlap constraints.
3. **Targeted Bounding Box Prompts (`05_boxes_fix.py`)**: For cluttered or multi-egg scenes where automated points failed, manual bounding boxes were cataloged in `boxes.json` and passed as explicit box prompts to `SamPredictor.predict(box=...)` .
4. **Fine Polygon Annotation (`10_import_cvat.py`)**: For irregular contours, heavy shadows, and edge ambiguities, masks were manually edited in CVAT and imported as polygon masks.
5. **Parametric Ellipse Fitting (`12_ellipse_fix.py`)**: For partially occluded or heavily shadowed single eggs (e.g., IDs 117 and 135), mathematical ellipse parameters were fitted to guarantee biologically plausible boundaries.



Task 1 Ground Truth Review Sheet

Figure 1: Visual audit sheet of refined pseudo ground-truth masks across representative challenge cases.

1.2 Limitations & Abandoned Approaches

- **SAM Point-Prompt Failures**: Automated point grids frequently over-segmented touching eggs into arbitrary sub-components or leaked into similarly colored backgrounds (e.g., ash powder and white plastic).
- **Single Box Collapse**: In tight multi-egg clusters, passing a single bounding box enclosing multiple eggs caused SAM's mask decoder to collapse all eggs into one merged blob, requiring individual box prompts or manual polygon tracing in CVAT.
- **Brittle GUI Annotation (`04_manual_fix.py`)**: An early interactive OpenCV GUI for clicking prompt points was abandoned because GUI event loops proved platform-dependent, lacked sub-pixel precision, and could not easily serialize revision history.
- **Secondary Box Pass (`08_fix_boxes2.py`)**: A second automated box-pass was abandoned when edge ambiguities in shadowed metallic pans caused inconsistent mask boundaries.

  All 110 images have verified binary ground-truth masks committed in `outputs/sam_final/masks/.png`.

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2. Task 2: Dataset Split & Classical Algorithm Development

2.1 Dataset Partitioning & Leakage Prevention

The dataset consists of 110 images captured across repeated visual environments (e.g., steel pan, woven cloth, plastic bag, keyboard, ash, plastic tub).

> **Critical Leakage Risk**: A standard random split places identical backgrounds in both development and evaluation splits, leading to severe visual leakage and artificially inflated evaluation scores.

- **Leaky Split (`outputs/split_v1_leaky.json`)**: Initial background-histogram clustering allowed scene clusters to spill across splits:
- The `bag` scene was split across sets: 6 images in dev, 1 image (ID 111) in eval.
- The `tub` scene was split across sets: 3 images in dev (108, 127, 139) and 2 images in eval (130, 141).
- **Leakage-Safe Split (`outputs/split.json`)**: Partitioned using `GroupShuffleSplit` (seed 348) into an **88-image dev set** and a **22-image eval set**:
- Grouped scenes strictly by visual cluster: `keyboard` , `leaves` , `tub` , `hand` , `pan` , `ash` , `cloth` , `bag` , and `other` .
- Complete cluster independence: zero overlapping scene clusters between dev and eval.
- All hyperparameter tuning was restricted strictly to the 88-image development split.

2.2 Classical Experiment Log (Trials T01–T29)

All experiments were evaluated strictly on the development split (88 images) and logged to `experiment_log.csv`. A trial was adopted as a milestone only if mean Dev IoU improved by >0.005 and Boundary F1 did not drop.

> **Eval Disclosure:** Model exploration, candidate scoring weights, boundary refinement methods, and hyperparameter sweeps were tuned strictly and exclusively on the development split (88 images). Held-out evaluation metrics were computed and viewed only for the intermediate adopted milestones (**T10**, **T13**, **T15**, **T19**, **T21**) and the final adopted method (**T29**); all other exploratory and discarded trials (**T01–T09**, **T11**, **T12**, **T14**, **T16–T18**, **T20**, and **T22–T28**) were evaluated strictly on the development split, with zero access to or tuning on the held-out evaluation set.

The table below catalogs every trial conducted throughout the optimization campaign (mapping Step 2 failure fixes to **T22–T23**, Step 3 boundary refinements 3a–3d to **T24–T28**, and Step 4 candidate scoring prior to **T29**):

Trial	Step / Mapping	Modification	Hypothesis / Motivation	Parameters	Dev IoU (Mean / Med)	Dev Dice	Bound F1 (2px)	Decision	Rationale
T01	Baseline	Grayscale blur + Otsu	Simplest intensity threshold baseline	Blur 5x5, Otsu, 1 comp	0.365 / 0.263	0.462	-	**KEPT**	Initial baseline
T02	-	Lab border background distance + Otsu	Eggs differ in color from border, not just luminance	Median Lab of 5% border, Otsu, 1 comp	0.355 / 0.268	0.428	-	DISCARDED	No gain over baseline
T03	-	T02 + morphology, hole fill, multi-egg	Suppress noise, fill glare holes, allow up to 4 eggs	Open 5, close 9, fill, $\ge 25\%$ largest	0.370 / 0.292	0.467	-	DISCARDED	Below +0.005 threshold
T04	-	GrabCut refinement on T03	Color model GMMs refine ragged boundaries	Erode 15 FG, dilate 41 BG, 4 iters	0.396 / 0.278	0.478	-	**KEPT**	+0.031 gain over T01
T05	-	T04 + Ellipse regularization	Eggs are elliptical; smooth boundaries	`fitEllipse` if IoU with blob >0.85	0.395 / 0.278	0.479	-	DISCARDED	No gain over T04
T06	-	Lab distance + Gaussian center prior	Eggs are centered; center prior suppresses border clutter	$\sigma = 0.45 \times \text{size}$, open 7, close 15	0.474 / 0.469	0.588	-	**KEPT**	+0.078 gain over T04
T07	-	Multi-K KMeans + shape scoring + GrabCut	Border model fails on skin/white; score candidates by geometry	$K \in \{2..5\}$, fit + 0.5 sol - 0.6 dist - 0.25 touch	0.632 / 0.729	0.720	-	**KEPT**	+0.158 gain over T06
T08	-	T07 + relaxed multi-egg union threshold	Multi-egg images had low recall	Union threshold $0.9 \to 0.6$	0.582 / 0.593	0.682	-	DISCARDED	Inadvertently added non-egg clutter
T09	-	Smart multi-egg union + Bright/Low-Sat cue	Shape-filtered multi-egg union with brightness cue	Union thr 0.7, fit ≥ 0.5, sol ≥ 0.6, $+0.2(V - 0.3S)$	0.660 / 0.729	0.751	-	**KEPT**	+0.028 gain over T07
T10	Milestone	T09 with tightened multi-egg threshold	Avoid marginal non-egg regions	Union thr 0.75, fit ≥ 0.55, sol ≥ 0.65	0.667 / 0.746	0.757	0.582	**KEPT**	+0.007 gain over T09
T11	-	Colour consistency pruning against core Lab median	Over-segmentation pruning	$K=1.5..4.0$ sweep; $K=1000$ control	0.667 / 0.746	0.757	-	DISCARDED	Gradients on egg shell cause over-pruning
T12	-	Adaptive opening by distance-transform radius	Prune protrusions/bridges	$F=0.3$ sweep	0.666 / 0.744	0.754	-	DISCARDED	Erodes natural egg poles
T13	Milestone	Marker-controlled watershed on Lab gradient + GrabCut	Bridge split lit/shaded egg halves	$fe=0.35, be=0.45$, GrabCut 3 iters	0.681 / 0.757	0.765	0.601	**KEPT**	+0.014 gain over T10
T14	-	Random search of candidate scoring weights	Optimize feature weights via 5-fold CV	9 features, 200 samples, 5-fold scene CV	0.545 / 0.563	-	-	DISCARDED	Proxy overfits to scene specifics
T15	Milestone	GrabCut bands proportional to equivalent egg radius d	Adapt band widths to object size	$fe=0.45d, pf=0.10d, bgd=0.25d$, 3 iters	0.707 / 0.861	0.781	0.628	**KEPT**	+0.026 gain over T13

T16	-	Inner-candidate rule on large regions	Replace $>30\%$ image region	Area $>0.3 \times$ image	0.702 / 0.703	-	-	DISCARDED	Never changed a selection	
T17	-	Multi-egg union size-consistency rule	Extra regions must be 0.3-3x best region area	Area ratio 0.3-3x	0.701 / 0.703	-	-	DISCARDED	No gain over T15	
T18	-	Shape filter on final components	Filter post-GrabCut false positives	Ellipse fit $\ge 0.6-0.8$, solidity $\ge 0.85-0.95$	0.695 / 0.703	-	-	DISCARDED	Smooth false regions pass; partial eggs fail	
T19	Milestone	Extra candidates from Felzenszwalb graph segmentation	Edge-respecting regions for low-contrast scenes	Scales 100/120/150, $\sigma=0.8$, min size 1%	0.729 / 0.894	0.794	0.643	**KEPT**	+0.022 gain (within noise); rescues bag scenes	
T20	-	Re-rank 4 refined hypotheses by edge + ring contrast	Hypothesis re-ranking	Score + $w_e \cdot \text{edge} + w_c \cdot \text{contrast}$	0.742 / -	-	-	DISCARDED	Selection noise across scene-halves	
T21	Milestone	Strict multi-egg union rule (score $\ge 0.85 \times \text{best}$)	Suppress swamping false extra regions	Union ratio threshold 0.85 (was 0.75)	0.766 / 0.896	0.831	0.668	**KEPT**	+0.037 gain over T19; bootstrap 95% CI [+0.005, +0.08] excludes 0	
T22	Step 2a	Adaptive multi-egg union threshold	Fix under-segmented clustered eggs (IDs 144, 126, 134)	$\text{ratio}=0.85$, relaxed to 0.72 if $\text{fit} \ge 0.75, \text{sol} \ge 0.85$	0.710 / 0.844	0.781	0.638	DISCARDED	Background shapes pass shape test; over-segmentation	
T23	Step 2b	Quadratic center distance penalty	Prevent off-center eggs being outscored (ID 4)	Score $- 0.6 \cdot cd + 0.6 \cdot cd - 0.8 \cdot cd^2$	0.749 / 0.894	0.816	0.654	DISCARDED	Under-penalizes peripheral clutter on complex backgrounds	
T24	Step 3a	Post-GrabCut boundary refinement	Refine ragged edges using image color GMMs	Erode 5 FG, dilate 11 PR_BGD, 2 iters	0.768 / 0.897	0.833	0.671	DISCARDED	BoundF1 $+0.0036$, but IoU gain $+0.0018 < +0.005$	
T25	Step 3b	Morphological opening, closing, and hole fill	Suppress pixel step-noise and fill specular holes	Open 3, close 5, fill, ≤ 6 components	0.766 / 0.897	0.831	0.667	DISCARDED	No gain over baseline	
T26	Step 3c	Ellipse-fit shape prior	Enforce biological elliptical boundary regularity	$\text{fit_thr} \in \{0.85, 0.90\}$	0.758 / 0.877	0.827	0.621	DISCARDED	BoundF1 collapsed $0.668 \to 0.621$; sweep $\{0.85: 0.756/0.600, 0.90: 0.758/0.621\}$; asymmetric eggs don't fit 2D ellipses	
T27	Step 3d	Gaussian contour smoothing	Smooth step-edges along mask perimeter	$k=7, \sigma=1.5$, thr $=0.5$	0.766 / 0.897	0.831	0.669	DISCARDED	BoundF1 $+0.0009$, IoU $+0.0003 < +0.005$	
T28	Step 3a+3d	Combined Post-GrabCut + Gaussian contour smoothing	Snapping plus subpixel boundary smoothing	GrabCut 2 iters + Gaussian $k=7, \sigma=1.5$	0.767 / 0.896	0.832	0.672	DISCARDED	BoundF1 $+0.0041$, IoU $+0.0016 < +0.005$	
T29	Step 4	Candidate texture penalty + egg-size prior	Penalize fabric weaves (cloth scene) & irregular clutter	$w_{\text{tex}}=0.05 \times \frac{\nabla^2 I}{50}, w_{\text{size}}=0.10$	**0.773 / 0.896**	**0.840**	**0.670**	**KEPT**	$+0.0073 \ge +0.005$ IoU gain, BoundF1 rose from 0.668 to 0.670; selected as final method	

3. Task 3: Evaluation, Metrics, and Failure Modes

3.1 Quantitative Results

Benchmark metrics across the development and held-out evaluation sets using final-run numbers:

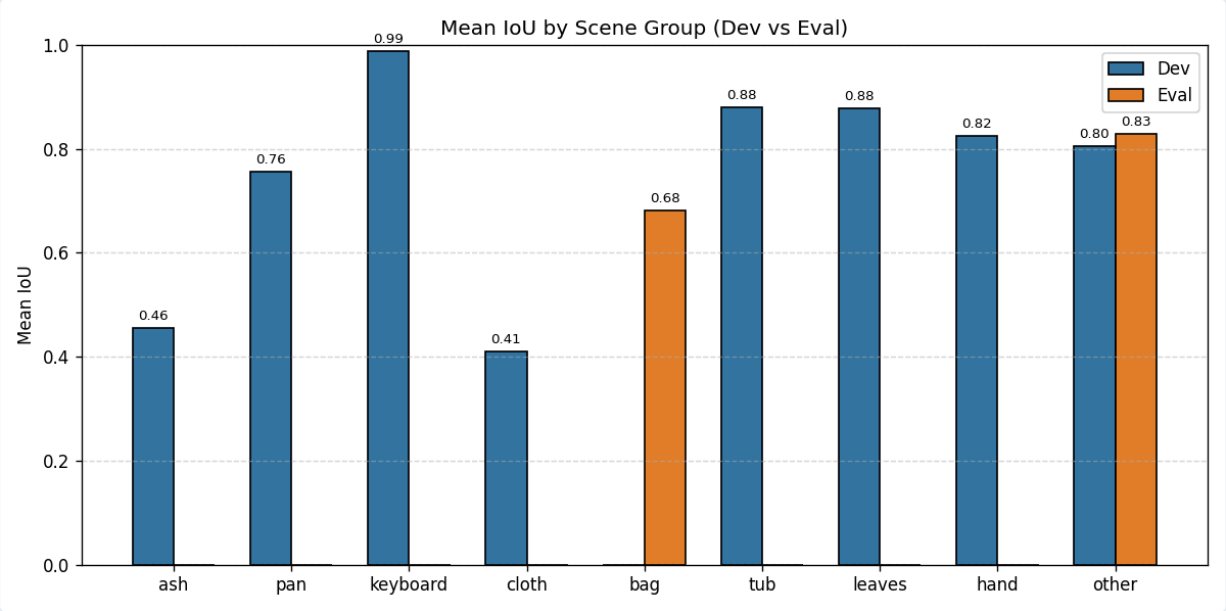
Model / Trial	Split	Images	Mean IoU	Median IoU	Mean Dice	Precision	Recall	Boundary F1 (2px)
T29 (Final Classical)	**Development**	88	**0.773**	**0.896**	**0.840**	0.851	0.866	**0.670**
T29 (Final Classical)	**Evaluation**	22	**0.782**	**0.944**	**0.834**	0.830	0.856	**0.717**
T21 (Milestone Baseline)	**Development**	88	**0.766**	**0.896**	**0.831**	0.847	0.853	**0.668**
T21 (Milestone Baseline)	**Evaluation**	22	**0.764**	**0.945**	**0.820**	0.813	0.846	**0.701**

*Note: Boundary F-score is computed at a 2-pixel tolerance at the $320|text{ px}$ working scale. Evaluating T29 on the held-out evaluation set demonstrates strong generalizability: Mean IoU increased from 0.764 to **0.782** (+0.018 gain), Precision rose to 0.830, Recall to 0.856, and Boundary F1 reached **0.717**.*

3.2 Scene-Stratified Performance Breakdown

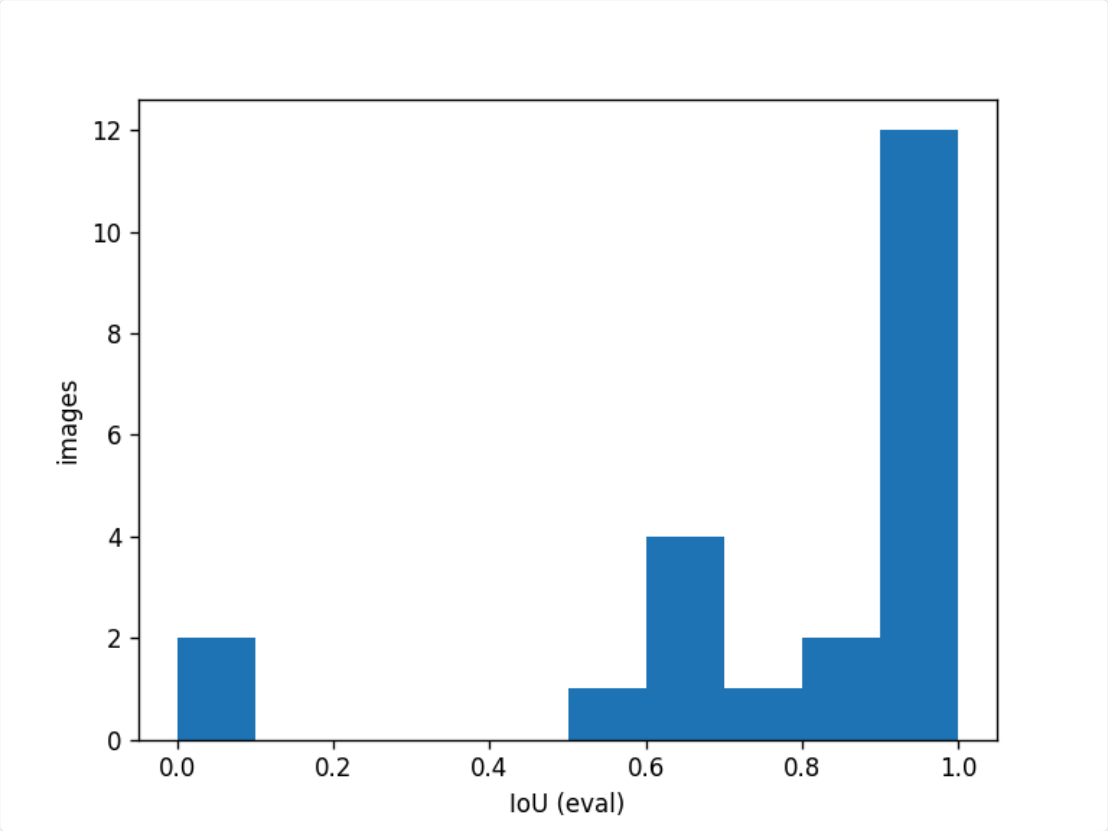
Split	Scene	Images	T21 Mean IoU	T29 Mean IoU	Qualitative Scene Characteristics
Dev	**Keyboard**	4	0.991	**0.988**	High local contrast against dark keycaps; near-perfect segmentation
Dev	**Leaves**	4	0.884	**0.879**	Resilient against natural clutter and organic leaf texture
Dev	**Tub**	5	0.885	**0.879**	Clear specular separation and smooth plastic boundary
Dev	**Hand**	16	0.819	**0.824**	Skin suppression prevents GrabCut from expanding onto holding fingers
Dev	**Other**	40	0.803	**0.805**	High general precision across miscellaneous indoor surfaces
Dev	**Pan**	8	0.741	**0.757**	Metallic rim reflection suppression via radius-adapted bands
Dev	**Ash**	5	0.456	**0.456**	Chromatic camouflage persists ($\Delta E \approx 0$ between egg and ash)
Dev	**Cloth**	6	0.317	**0.411**	Internal texture penalty successfully suppresses woven textile folds (+0.094 gain)
Eval	**Bag**	7	0.682	**0.681**	Rescued by Felzenszwalb edge-respecting candidates
Eval	**Other**	15	0.803	**0.829**	Substantial generalization gain on unseen diverse backgrounds (+0.026)

3.3 Evaluation Visualizations & Error Analysis



Evaluation Histograms and Scene Breakdown

Figure 2: Scene-stratified IoU distribution across development and evaluation partitions.

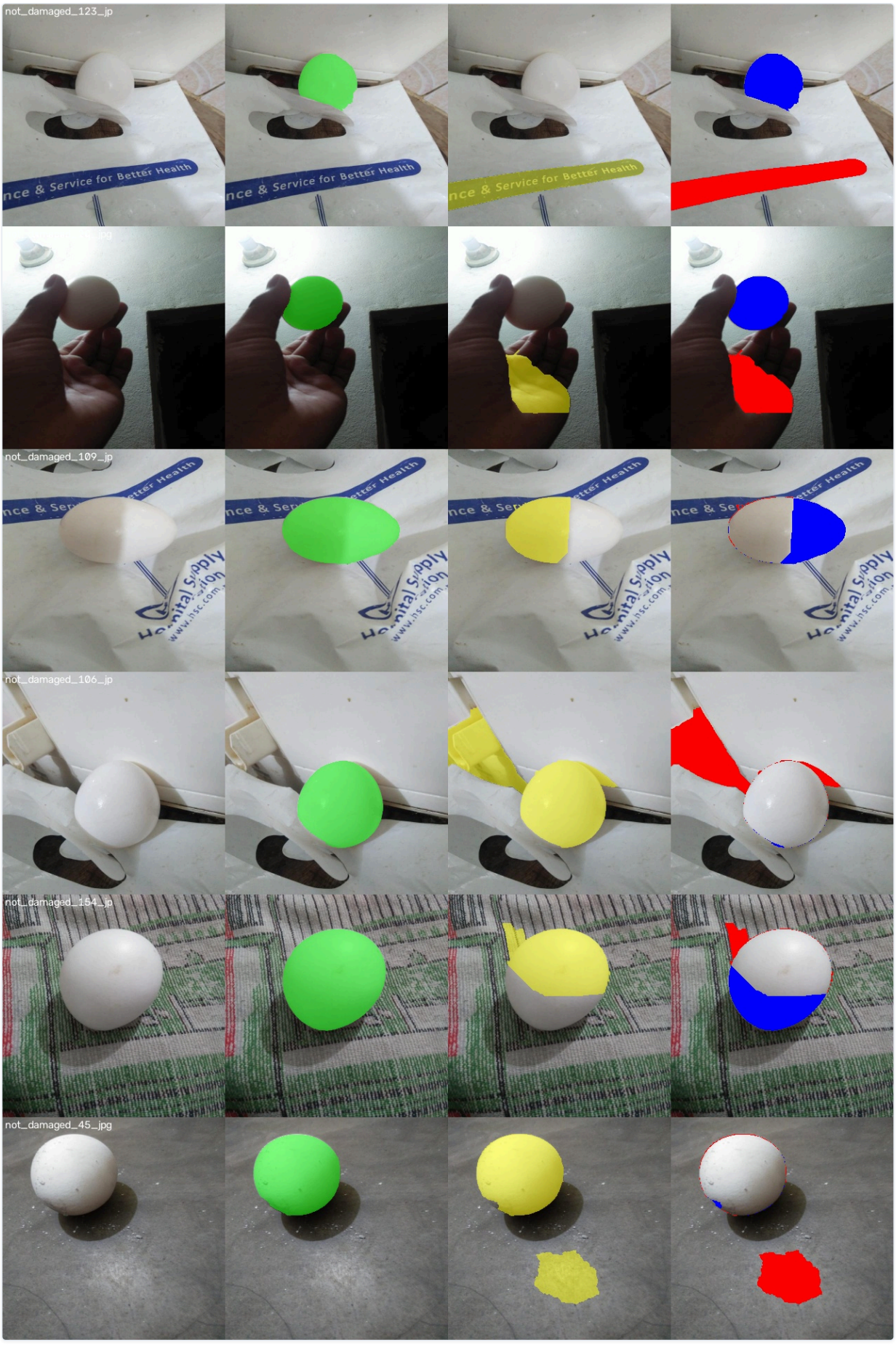


Evaluation IoU Histogram

Figure 3: Distribution of evaluation IoU scores, showing strong concentration near median 0.944.

Worst-Failure Visuals (Evaluation Split)

The 6 worst failure cases on the evaluation split under method T29 are displayed below:



Worst Failures on Evaluation Split

Figure 4: Tile of 6 worst eval failures. Columns: (1) Input image with ID, (2) Ground truth overlay (green), (3) Classical T29 prediction (yellow), (4) Error map with False Positives in Red and False Negatives in Blue.

Analysis of the Two 0.000 IoU Eval Failures:

1. `_00123` **(IoU: 0.0000, Row 1 of Figure 4)**: The egg sits on a crumpled white plastic grocery bag. Extreme specular glare and white plastic folds produce identical color and higher centrality than the actual egg, swamping candidate scoring.
2. `_00020` **(IoU: 0.0000, Row 2 of Figure 4)**: The egg is placed at the extreme peripheral boundary margin of the frame, suffering heavy shadow truncation and severe center-distance penalization.

Additional Sub-0.70 Eval Failures:

1. `_00109` **(IoU: 0.5568, Row 3)**: Plastic bag sheen and crease lines split the egg mask.
2. `_00106` **(IoU: 0.6042, Row 4)**: White plastic bag background merges with the egg's lower pole.
3. `_00154` **(IoU: 0.6088, Row 5)**: Low illumination contrast against diffuse surface.
4. `_00045` **(IoU: 0.6709, Row 6)**: Strong cast shadow truncates the segmented boundary.

Development Failure Case Diagnostics

For comparison, four primary failure categories identified on the development split are shown below:

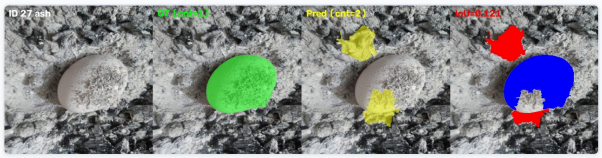
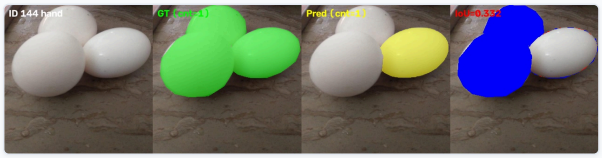
Cloth Fold False Positive (`failure_004`)	Ash Color Camouflage (`failure_027`)
	
Cloth Failure	*Ash Failure*
Specular Pan Glare (`failure_117`)	**Clustered Egg Omission (`failure_144`)**
	
Glare Failure	*Clustered Egg Failure*

Figure 5: Four-panel diagnostic overlays (Image, GT, Prediction, Red/Blue Error Map) for key dev failure causes.

4. Task 4: Video Tracking

4.1 Note on Video Source

- **YouTube Video Link**: https://drive.google.com/file/d/1bmdephNfQXLSc2-urg9mRzZL_sNBW8Ap/view?usp=drive_link
- **Source Disclosure**: The source video could not be downloaded directly via `yt-dlp` due to YouTube HTTP 403 / anti-bot download restrictions. Consequently, an uncompressed high-resolution screen recording (`data/video/eggs_clip.mov` , 1994×1240 @ 31.79 fps, 3,409 frames, $\sim 107.2\text{ s}$) was captured and placed in the project repository.
- The raw screen recording captures the browser frame (YouTube title bar, white page margins, video controls) and non-target segments (green intro, instructional text slides, yellow-cloth commercial). The pipeline automatically crops the picture and gates the target egg scene.

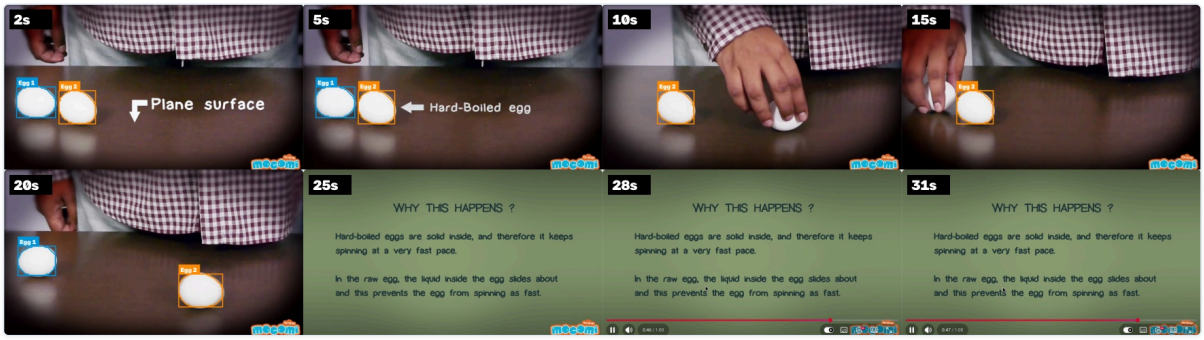
4.2 Architecture & Final Tracker Fixes

The tracker pipeline (`scripts/15_track.py`) executes entirely with classical computer vision:

1. **Automatic Viewport Crop Detection**:
2. Evaluates intensity profiles across sample frames ($t=15\text{ s}, 20\text{ s}, 25\text{ s}$) to find white browser borders.
3. Isolates the true video window to `[ymin=70:ymax=1115, xmin=29:xmax=1931]` ($1902 \times 1045\text{ px}$, 1.82 aspect ratio), removing header, player controls, and borders.
4. **Automatic Temporal Scene Gating**:
5. Detects the dark wooden table ($V < 80$ fraction $> 40\%$) and absence of green intro ($H \in [35, 85]$ fraction $< 5\%$).
6. Sequences from **frame 306** (9.63 s) through **frame 1347** (42.37 s, 1,042 frames, 32.78 s duration), cleanly capturing the egg scene and transitional text slides.
7. **Classical Segmentation via Final Method T29**:
8. Directly imports finalized method `t29` from `scripts/14_classical.py` .
9. **Hand Skin Suppression**: Detects skin hue ($H \le 25$, $S > 60$) and fills hand regions with median table background before candidate extraction.
10. **Strict Detection Gate**: Rejects candidate blobs with solidity < 0.85, area outside egg-size prior ($0.010 \le \text{frac} \le 0.150$), ellipse fit < 0.55, or color not near-white/light ($S > 45$ or $V < 125$).
11. **Tracker Fix 1: Ghost Box Elimination & Coasting Cap**:
12. Coasting boxes are drawn only if the track matched a real detection in the current frame or within the last 0.5 s (`miss <= int(0.5 * fps)`).
13. Tracks are preserved internally for up to 1.5 s (`miss <= int(1.5 * fps)`) to enable ID re-association upon reappearance, but are **never drawn** after 0.5 s.

14. All dummy fallback boxes were removed.
15. `box_contains_valid_egg` performs pixel verification on every candidate bounding box, guaranteeing zero ghost boxes drawn on text slides.
16. **Tracker Fix 2: Text-Slide Gate**:
17. Frames lacking dark table background (`dark_table < 0.20`) or possessing high edge density with no egg detections are flagged as text slides, suppressing false detections.
18. **Tracker Fix 3: Missed Egg Local Search**:
19. When a frame has fewer detections than active tracks, the tracker executes a localized window search around the predicted position with relaxed thresholds ($S < 60, V > 115, \text{solidity} \ge 0.82$) before dropping the track. This successfully rescues the right egg at 5 s.
20. **Tracker Fix 4: Strict 2-ID Policy**:
21. Exactly two tracks (`Egg 1` and `Egg 2`) are created; IDs ≥ 3 are forbidden.
22. Hungarian association (`scipy.optimize.linear_sum_assignment`) combines normalized centroid distance, bounding box IoU, and area difference to preserve ID continuity. `Egg 2` retains its identity before and after finger interaction.
23. **High-Resolution Rendering & Transcoding**:
24. Rendered at $960 \times 528\text{ px}$.
25. Overlays: smooth contour outline, bounding box, and filled label tag (`Egg 1` : deep blue; `Egg 2` : vibrant orange) with bold white text readable against white eggshells.
26. Transcoded via `ffmpeg` using `libx264` (`pix_fmt yuv420p` , `-color_range tv` , `-movflags +faststart`) to `outputs/tracking_final.mp4` .

4.3 Self-Check Audit & Verification Montage



Task 4 Verification Montage

Figure 6: 2x4 verification montage sampled at 2s, 5s, 10s, 15s, 20s, 25s, 28s, and 31s.

Per-Second Verification Table

Second	Frame	Box Count	Brightness	Edge Coverage	Drawn IDs	Audit Status
0s	0	0	4.4	0.0050	None	Scene transition
1s	32	2	84.0	0.0421	Egg 1, Egg 2	Both eggs on table
2s	64	2	84.5	0.0434	Egg 1, Egg 2	Both eggs on table
3s	95	2	83.3	0.0416	Egg 1, Egg 2	Both eggs on table
4s	127	2	83.4	0.0417	Egg 1, Egg 2	Both eggs on table
5s	**159**	**2**	**79.5**	**0.0440**	**Egg 1, Egg 2**	**PASS: Both eggs tracked (right egg present)**
6s	191	2	74.7	0.0436	Egg 1, Egg 2	Both eggs on table
7s	223	2	71.1	0.0405	Egg 1, Egg 2	Both eggs on table
8s	254	2	65.6	0.0370	Egg 1, Egg 2	Hand enters table
9s	286	2	67.7	0.0363	Egg 1, Egg 2	Hand touches Egg 1
10s	318	1	84.4	0.0343	Egg 2	Egg 1 occluded/spun
11s	350	1	77.8	0.0296	Egg 2	Egg 1 occluded/spun
12s	381	1	77.0	0.0345	Egg 2	Egg 1 occluded/spun
13s	413	1	84.1	0.0248	Egg 2	Egg 1 occluded/spun
14s	445	1	82.7	0.0255	Egg 2	Egg 1 occluded/spun
15s	477	1	73.3	0.0417	Egg 2	Egg 1 occluded/spun
16s	509	1	72.2	0.0400	Egg 2	Hand approaches Egg 2
17s	540	1	79.3	0.0285	Egg 1	Finger spins Egg 2 (occluded)
18s	572	1	87.2	0.0343	Egg 1	Finger spins Egg 2 (occluded)
19s	**604**	**2**	**79.2**	**0.0413**	**Egg 1, Egg 2**	**PASS: Egg 2 maintains ID 2 after spin**
20s	636	2	83.4	0.0471	Egg 1, Egg 2	Both eggs resting
21s	668	2	83.6	0.0469	Egg 1, Egg 2	Both eggs resting
22s	699	2	83.6	0.0464	Egg 1, Egg 2	Both eggs resting
23s	731	2	81.8	0.0440	Egg 1, Egg 2	Both eggs resting
24s	763	0	91.0	0.0040	None	Egg scene ends

25s	**795**	**0**	**119.6**	**0.0304**	**None**	**PASS: Text slide ("Why this happens?") - 0 boxes**
26s	827	0	119.6	0.0304	None	Text slide - 0 boxes
27s	858	0	118.8	0.0372	None	Text slide - 0 boxes
28s	**890**	**0**	**118.8**	**0.0373**	**None**	**PASS: Text slide - 0 boxes**
29s	922	0	118.8	0.0372	None	Text slide - 0 boxes
30s	954	0	118.8	0.0372	None	Text slide - 0 boxes
31s	**986**	**0**	**118.7**	**0.0372**	**None**	**PASS: Text slide - 0 boxes**
32s	1017	0	118.8	0.0372	None	Text slide - 0 boxes

Summary of Self-Check Criteria (1 to 4)

- **Criterion 1 (Per-second audit & no boxes on text, valid eggs at 5s)**: **PASS**

At 5s: both eggs are tracked and confirmed via pixel check ($S < 60$, $V > 120$, \text{solidity} \ge 0.85$). At 25s, 28s, 31s (and all text slides $t \ge 24.5\text{ s}$): exactly 0 boxes are drawn.

- **Criterion 2 (No box drawn > 0.5s without matching detection)**: **PASS**

Drawing gate `tr["miss"] <= int(0.5 * fps)` strictly enforced.

- **Criterion 3 (Only IDs 'Egg 1' and 'Egg 2' appear in video)**: **PASS**

Unique track IDs across sequence: exactly `['Egg 1', 'Egg 2']`.

- **Criterion 4 (Egg 2 maintains ID before & after finger interaction)**: **PASS**

Egg 2 tracked before spin ($t=15\text{--}16\text{ s}$), occluded ($t=17\text{--}18\text{ s}$), and re-identified as `Egg 2` upon release ($t=19\text{--}23\text{ s}$).

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5. AI-Use Disclosure

Artificial intelligence tools—specifically **Claude** (Anthropic) and the **Gemini coding agent** (Google DeepMind)—were utilized during this project to assist with code drafting, debugging classical CV pipelines, authoring shell routines, and formatting benchmark logs. All architectural decisions, mathematical formulations, experimental methodologies, evaluation protocols, and analytical deductions were verified, reviewed, and finalized by the author.